Deep Learning Regression

Predicting wine quality from physicochemical properties using neural networks

Quang Minh Nguyen
https://github.com/Minh111104/Deep-learning-with-Wine-Quality-Dataset

Overview

- 2 datasets:
  - *winequality-red.csv*: 1,599 instances
  - *winequality-white.csv*: 4,898 instances

- Input/Features:
  - 11 physicochemical (*fixed_acidity*, *volatile_acidity*, *citric_acid*, etc.)
  - *wine_type* (0=red, 1=white)

- Output/Target:
  - *quality*: score between 0 and 10

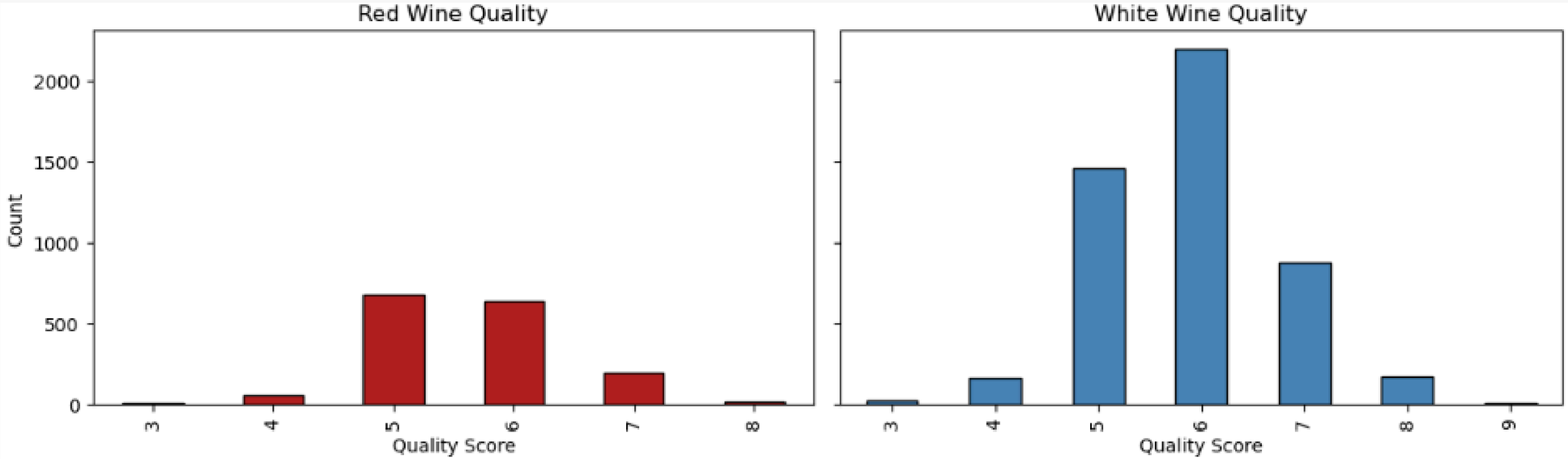
- Libraries used:
  - pandas, numpy, matplotlib, scikit-learn, tensorflow

Data Exploratory & Preprocessing

- No missing values (6,497 rows × 12 columns)
- 12 features: 11 physicochemical + *wine_type*
- Train/Validation/Test split: 70/15/15
- Scale features using *StandardScaler*

```
Train:      (4547, 12)
Validation: (975, 12)
Test:       (975, 12)
```

- Distribution of quality scores by wine type:



Build Neural Network

- Used Keras Sequential model
- Linear output activation for continuous regression output

- Architecture:
  - Input(12) → Dense(64, ReLU) → Dense(32, ReLU) → Dense(16, ReLU) → Dense(1, linear)

Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 64)	832
dense_1 (Dense)	(None, 32)	2,080
dense_2 (Dense)	(None, 16)	528
dense_3 (Dense)	(None, 1)	17

Total params: 3,457 (13.50 KB)

Trainable params: 3,457 (13.50 KB)

Non-trainable params: 0 (0.00 B)

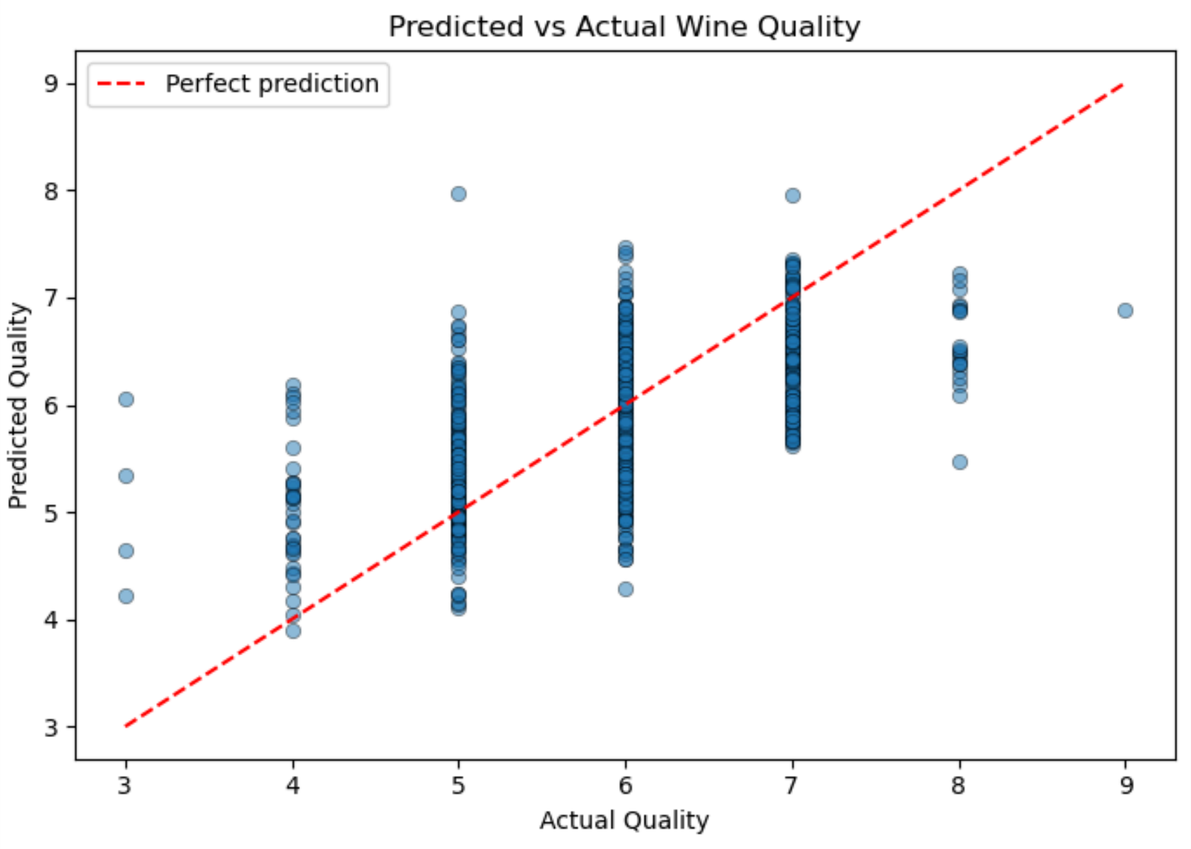
Compile & Train

- Optimizer: *Adam*
- Loss: *MSE* (Mean Squared Error)
- Metric: *MAE* (Mean Absolute Error)
- *EarlyStopping* with *patience=10*, restores best weights
- Max 200 epochs, batch size 32

Evaluate on Test Set

- Evaluated on held-out test set: MSE, MAE, RMSE

```
Test MSE: 0.4755
Test MAE: 0.5397
Test RMSE: 0.6895
```



Key Takeaways

- Feature scaling is critical: *StandardScaler* significantly improves neural network convergence
- Linear output activation is required for regression: sigmoid/softmax would constrain predictions to a fixed range
- *EarlyStopping* prevents overfitting: training stopped early and restored the best weights automatically
- More data gives the model more to learn

Thank You!

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